

The Zimmerman Theory of Gravity

The galaxy acceleration scale is set by the cosmological constant: $a_0 = c^2 \sqrt{\Lambda/32\pi}$

Comprehensive edition — the law, its de Sitter-vacuum foundation, its redshift evolution, and the measurements that decide it

Author: Carl P. Zimmerman (Briar Creek Tech) · correspondence: carl@briarcreektch.com **Version:** Comprehensive edition v2 — 2026-06-06 (supersedes the draft-1 whitepaper) **Code & data:** <https://github.com/carlzimmerman/zimmerman-formula> — every quantitative claim links to a runnable Python script in `real_research/` and a public dataset. **Status:** A falsifiable proposal offered for independent testing. Nothing here requires trusting the author — clone the repo, pull the public data, and re-run it.

Abstract

The dynamics of galaxies require either unseen matter or a modification of gravity below a characteristic acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$. It has long been noted (Milgrom 1983, 1999; Famaey & McGaugh 2012) that this scale numerically coincides with $c \sqrt{\Lambda}$ and with cH_0 — a coincidence Λ CDM treats as accidental. The **Zimmerman Theory of Gravity** proposes that the coincidence is causal: the acceleration scale is **set by the cosmological constant** (the dark-energy density of the vacuum),

$$a_0 = c^2 \sqrt{\Lambda/32\pi} = (c/2) \sqrt{G\rho_\Lambda} = cH_0 / Z, Z = 2 \sqrt{8\pi/3} = \sqrt{32\pi/3} = 5.789, a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$$

evaluated on the dark-energy density alone ($\rho_\Lambda = \Omega_\Lambda \rho_{\text{crit}}$). Beyond the empirical confrontation, this comprehensive edition assembles the full **theoretical foundation** the framework has earned, layer by layer with honest labels: (i) the deep-MOND **shape** $g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + g_{\text{bar}} \cdot a_0}$ is **derived** — it is Milgrom’s (1999) de Sitter–Unruh modified-inertia form, overdetermined across three routes, not a fitted interpolation; (ii) the **existence** of a_0 is supplied **volume-law-free** by that modified-inertia route (the kinematic de Sitter–Unruh temperature of Deser–Levin 1997), so the framework’s deepest vulnerability is properly the *covariant completion*, not “does a_0 exist”; (iii) the deep-MOND **sign** (enhancement, not screening) is **forced for galaxy-scale probes**, conditional on the Narovlansky–Verlinde de Sitter/DSSYK dictionary, by a computed matter-chord kernel that *also* predicts MOND’s empirical failure in clusters; (iv) the **value of Λ** itself is welded to a_0 as the two ends of one Cohen–Kaplan–Nelson UV–IR $\sqrt{\Lambda}$ ladder ($\rho_{\text{obs}} = (3/8\pi)M_P^2 H^2$, the CKN

bound exactly saturated); and (v) the framework’s distinctive **evolving** dark energy is **string-swampland-compatible** precisely where a static- Λ Λ CDM is swampland-forbidden. We are equally explicit about the limits: the $O(1)$ coefficient (32π) is an undetermined posit (foreclosed across a six-route assault); the value-of- Λ welding *relocates* rather than *solves* the cosmological-constant problem; this is a theory of **gravity and the dark sector, not a theory of everything** (the Standard Model is untouched; §9, §13). We confront the law with five independent datasets — at its own a_0 and a single $\Upsilon \approx 0.70$ the Radial Acceleration, Baryonic Tully–Fisher, and deep-MOND mass-discrepancy relations agree to 8%; the **rising** $a_0 \propto cH$ rival is excluded ($\Delta\chi^2 \approx 49$); the Λ CDM-impossible External Field Effect leans the predicted way ($\sim 1.4\sigma$, data-limited) — and we specify the precise empirical thresholds (§12) that would promote the proposal from candidate to law. The single decisive measurement, deep-MOND kinematics at $z \approx 3$ returning $a_0(z=3) = 0.74 a_0(0)$, is reachable with ELT-class spectroscopy this decade.

1. Introduction

Spiral galaxies rotate too fast for their visible mass. The two responses are (a) cold dark matter halos, and (b) a breakdown of Newtonian dynamics below an acceleration scale a_0 (Modified Newtonian Dynamics; Milgrom 1983). MOND is empirically remarkable at galaxy scales: it predicts rotation curves from the baryons alone with a single universal constant a_0 , and it anticipated the Radial Acceleration Relation and the Baryonic Tully–Fisher Relation decades before they were measured (McGaugh, Lelli & Schombert 2016; Lelli et al. 2016, 2019).

The unexplained fact at the centre of this paper is the **value** of a_0 . Numerically, $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2} \approx cH_0/2\pi \approx c^2 \sqrt{\Lambda} \times O(1)$.

In Λ CDM this is a coincidence: there is no acceleration constant in the theory at all (galaxies are dark-matter halos), and the equality of a galaxy-dynamics scale with a cosmological one is accidental. The Zimmerman Theory takes the coincidence literally: **the vacuum sets the scale**. This is in the spirit of the dark-energy/ a_0 link proposed by Limbach, Psaltis & Özel (2008), made specific here by a fixed coefficient and, crucially, by a definite **redshift evolution** tied to the measured dark-energy equation of state.

On the word “theory” / “law.” We use “law” in the sense of an empirical regularity with a parameter-free form whose constants are fixed externally — the standard a relation must clear to earn the title (§12). We use “theory” in the sense that MOND, TeVeS and emergent gravity are theories: a modified-gravity framework with a derived scale and falsifiable predictions. The empirical content currently stands at the level of a strong **candidate** law; its covariant, ghost-free field-theoretic completion is an open problem (§10), exactly as it is for modified gravity in general. We flag every weak point honestly throughout — a theory

worth testing is one whose vulnerabilities are named, and this edition names them at the level of the individual derivation step (§4).

What this edition adds over draft 1. The first whitepaper assembled the empirical case. This comprehensive edition folds in the full theoretical-foundation program: the layer-by-layer derivation ledger (§4), the modified-inertia reconciliation of a_0 's existence (§4.2), the computed deep-MOND sign and the galaxy/cluster split (§4.4), the cosmic seesaw for the value of Λ (§5), the string-swampland compatibility and the honest limits of unification (§9), and an explicit retraction of the numerological material that must never be attached to the genuine result (§13).

2. Statement of the law

2.1 The acceleration scale. The single new constant is

$$a_0 = c^2 \sqrt{(\Lambda/32\pi)} = (c/2) \sqrt{(G\rho_\Lambda)} = \sqrt{(8\pi G\rho_\Lambda/3)} \cdot (c/Z) = cH_\Lambda/Z,$$

with $Z = 2 \sqrt{(8\pi/3)} = \sqrt{(32\pi/3)} = 5.789$, $H_\Lambda = \sqrt{(\Lambda c^2/3)}$ the de Sitter (pure- Λ) expansion rate, and ρ_Λ **the dark-energy density alone** ($\rho_\Lambda = \Omega_\Lambda \rho_{\text{crit}}$; not the total matter+ Λ density). With the Planck/DESI values ($\Omega_\Lambda=0.685$, $H_0=67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Lambda=1.09 \times 10^{-52} \text{ m}^{-2}$):

$$a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}.$$

Footing matters. Evaluating the same formula on the **total** density (ρ_{crit} , equivalently cH_0 with the matter-inclusive Hubble rate) gives 1.13×10^{-10} — a different number with a different, monotonically **rising** redshift evolution. The law's footing is the **dark energy alone** ($\rho_{\text{DE}} / cH_\Lambda$); this choice is the source of the declining evolution (§2.4) and is essential. (Verified: `framework_a0_law_of_nature.py`, `coefficient_posit_attack.py`.)

2.2 The coefficient, honestly. The combination $c^2 \sqrt{\Lambda}$ is forced by dimensional analysis once the vacuum sets an acceleration; only the dimensionless $O(1)$ coefficient is a choice. The law posits $X = 32\pi$ in $a_0 = c^2 \sqrt{(\Lambda/X)}$. This factorizes as $32\pi = 4 \times 8\pi$: the 8π is GR-forced (it is the Einstein coupling, the same 8π that relates ρ_Λ to Λ), and the residual $4 = (1/2)^2$ is the surface-gravity $1/2$ of a collapse/free-fall horizon at radius $R_\star = \sqrt{(8\pi/3)} R_{\text{dS}}$, squared. De Sitter equipartition, Unruh, and holographic-equipartition routes give $X = 3$ or $X = 12\pi^2$ — **none forces 32π** . We therefore present 32π as a **motivated normalization posit on a GR-forced skeleton, not a theorem** (`coefficient_posit_attack.py`). A comprehensive six-route assault — de Sitter equipartition, Unruh, holographic equipartition, the CKN UV-IR seesaw, AeST, and DSSYK — **fails to derive it**: every route either reduces to an algebraic tautology (the Friedmann relation, true for any ρ) or

inherits a proven no-go (GEOMETRIC_CLOSURE_ASSAULT_2026-06-06.md). Crucially, **the coefficient is observationally moot** — it cancels identically in the only falsifiable content, the ratio $a_0(z)/a_0(0)$ (§2.4) — so the posit weakens no empirical claim. The coefficient is to this theory what the precise value of a_0 is to Milgrom’s MOND or the “6” is to Verlinde’s emergent gravity: an underived normalization the physics does not depend on.

2.3 The interpolation (the RAR shape). Observed and baryonic accelerations are related by

$$\mathbf{g}_{\text{obs}} = \sqrt{(\mathbf{g}_{\text{bar}})^2 + \mathbf{g}_{\text{bar}} \cdot \mathbf{a}_0},$$

the “simple” interpolation. This shape is **not extra freedom** — §4.3 shows it is the closed-form solution of de Sitter–Unruh modified inertia, identical to Milgrom’s (1999) independently-derived form. It gives a **parameter-free** prediction for the entire Radial Acceleration Relation (Fig. 1); fit to 175 SPARC galaxies it yields 0.105 dex scatter (`rar_emergent_discriminate.py`).

2.4 The evolution (the distinctive content). Because a_0 tracks the dark-energy density,

$$\mathbf{a}_0(z) = \mathbf{a}_0 \cdot \sqrt{(\rho_{\text{DE}}(z)/\rho_{\text{DE}}(0)), \rho_{\text{DE}}(z)/\rho_{\text{DE}}(0) = (1+z)^{-3(1+w_0+w_a)}} e^{-3w_a z/(1+z)} \text{ (CPL)}.$$

With the DESI DR2 equation of state ($w_0=-0.752$, $w_a=-0.86$):

z	$a_0(z)/a_0(0)$	note
0.0	1.000	today
0.4	1.062	the bump (+6%)
1.0	1.009	
2.0	0.862	declining
3.0	0.737	declining

This **non-monotonic, net-declining** $a_0(z)$ is the theory’s signature (Fig. 2). It differs qualitatively from regular MOND (constant a_0) and from Verlinde-type emergent gravity ($a_0 \propto cH(z)$, **rising** steeply with z), and Λ CDM has no a_0 to evolve. **The late-time decline is a prediction of evolving dark energy, not of a constant Λ** — the instantaneous w runs from -0.75 today to ≈ -1.4 by $z=3$, crossing -1 at $z \approx 0.41$ (which produces the +6% bump before the decline). A *true* cosmological constant ($w=-1$) gives $\rho_{\text{DE}} = \text{const} \rightarrow a_0 = \text{constant}$, no evolution at all. So the theory’s galaxy-scale prediction and DESI’s cosmological measurement are *locked together*: DESI sees evolving dark energy $\Leftrightarrow a_0$ declines. (Verified in every script; canonical in `a0z_clean_ledger.py`.)

3. Provenance and what is new

Honest scholarship strengthens the case. This law stands on a lineage: - **Milgrom (1983)** — MOND and the acceleration scale a_0 ; **Milgrom (1999, “MOND as a vacuum effect”)** — $a_0 \sim c \sqrt{\Lambda/3}$ from the de Sitter / Unruh vacuum, *and the modified-inertia interpolation this work derives in §4.3.* - **Deser & Levin (1997)** — the kinematic de Sitter–Unruh temperature $T(a) = (1/2\pi) \sqrt{a^2 + \Lambda/3}$, the global-embedding result on which the existence of a_0 rests (§4.2). - **Limbach, Psaltis & Özel (2008, arXiv:0809.2790)** — *the coupling itself*: they contrasted $a_0 \sim cH_0$ with $a_0 \sim \sqrt{8\pi G\rho_{\Lambda}/3}$, derived the implied **declining-into-the-past** $a_0(z)$, and found high- z Tully–Fisher systematics marginally favor the dark-energy-density coupling. The $\sqrt{\rho_{\text{DE}}}$ coupling and its decline are theirs; we do not claim them. - **Cohen, Kaplan & Nelson (1999)** — the UV–IR seesaw bound $\rho \lesssim M_{\text{P}}^2 H^2$ that the value-of- Λ welding (§5) saturates. - **McGaugh, Lelli & Schombert (2016); Lelli et al. (2016, 2019)** — the RAR and BTFR as tight empirical laws. **Famaey & McGaugh (2012)** — the review establishing the a_0 – Λ coincidence as a real puzzle.

Original to this work: (i) the **specific value** via the fixed coefficient $a_0 = c^2 \sqrt{\Lambda/32\pi}$ with the dark-energy (ρ_{DE}) footing and the collapse-horizon interpretation of the $O(1)$; (ii) the **DESI- $w_0 w_a$ evaluation** yielding the non-monotonic, declining $a_0(z)$ with the $z \approx 0.4$ bump; (iii) the **systematic five-dataset confrontation** (§6–8); (iv) the **assembled theoretical-foundation ledger** (§4) — in particular the modified-inertia reconciliation of a_0 ’s existence and the *computed* deep-MOND sign with its galaxy/cluster split; and (v) the explicit demonstration that the framework’s evolving dark energy is **swampland-compatible** where ΛCDM is not (§9). We claim the synthesis, the value’s evolution, the confrontation, and the foundation ledger — not the existence of MOND or of the coincidence.

4. Theoretical foundation — the derivation chain from the de Sitter vacuum

A modified-gravity proposal is only as strong as the honesty of its foundation. This section states, step by step and with a hard label on each, exactly what the de Sitter vacuum **forces**, what is **dimensional**, what is **posited**, and what is **contested**. The result is mixed and stated as such: a genuinely derived core (the deep-MOND shape; the galaxy-scale sign) sitting on a dimensional scale, a volume-law-free existence, a posited coefficient, and an unfinished covariant completion. (Full synthesis: `FIRST_PRINCIPLES_FOUNDATION_2026-06-06.md`, `FOUNDATIONS.md`.)

4.1 The chain, every step labeled.

Step	Claim	Status	Basis (verified)
0. Existence of an a_0 term	the vacuum produces a low-acceleration scale at all	✓ volume-law-FREE in modified inertia	the de Sitter–Unruh <i>temperature</i> (kinematic; Deser–Levin 1997) gives modified inertia $\mu(a)$; a_0 exists with no entropy assumption (§4.2)
1. Scale $a_0 \sim \frac{c}{\sqrt{\Lambda}}$	the magnitude is set by the horizon	DIMENSIONAL (robust)	$a_0 \sim c^2/R_{\text{dS}}$ is the unique acceleration from $\{c, R_{\text{dS}}\}$; forced for every emergent-gravity scheme
2. Form $a_0 \propto \sqrt{\rho_{\text{DE}}}$	the evolution	RESTATEMENT + a CHOICE	exponent 1/2 is an exact algebraic identity given $a_0 \propto 1/R_{\text{dS}}$; the <i>falsifiable</i> part is the choice of the <i>declining</i> (ρ_{DE}) branch (§2.4)
3. Shape $g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + g_{\text{bar}} \cdot a_0}$	the deep-MOND $\sqrt{}$ -law	✓ GENUINELY DERIVED	closed-form solution of the dS–Unruh inertia law = Milgrom 1999 eq.(9); over-determined across three routes (§4.3)
4. Response posit	inertia tracks the <i>excess</i> Unruh heat $\Delta T = T(a) - T(0)$	POSIT (load-bearing for step 3)	the $\sqrt{}$ -law needs the floor-subtraction; Milgrom himself flags it as not fully justified

Step	Claim	Status	Basis (verified)
5. Sign (enhancement, not screening)	gravity strengthens below a_0	FORCED FOR GALAXIES, given the N-V dictionary	computed DSSYK matter-chord kernel: galaxies → spectral center → MOND; clusters → edge → MOND fails (§4.4)
6. Number $Z = \sqrt{32\pi/3} = 5.789$	the exact $O(1)$ coefficient	$\times$ FORE- CLOSED POSIT	a six-route assault failed; $32\pi/3$ enters only as the <i>definition</i> of a_0 (§2.2)

The derivation chain from the de Sitter vacuum — every step labelled honestly

0 Existence of a_0	<i>volume-law-FREE (modified inertia)</i>
1 Scale $a_0 \sim c/\Lambda$	<i>DIMENSIONAL (robust)</i>
2 Form $a_0 \propto \sqrt{\rho_{DE}}$	<i>restatement + a falsifiable choice</i>
3 Shape $\sqrt{(g^2 + g \cdot a_0)}$	<i>DERIVED = Milgrom 1999</i>
4 Response (inertia$\leftarrow\Delta T$)	<i>POSIT (load-bearing)</i>
5 Sign (enhancement)	<i>FORCED for galaxies (given N-V)</i>
6 Number $Z=\sqrt{32\pi/3}$	<i>POSIT (foreclosed, but moot)</i>

DERIVED

FORCED (given a named premise)

volume-law-free existence

dimensional / restatement

POSIT

Figure 6. The derivation chain from the de Sitter vacuum, every step labelled honestly: the deep-MOND *shape* is genuinely derived (= Milgrom 1999), the galaxy-scale *sign* is forced given the Narovlansky–Verlinde dictionary, the *existence* of a_0 is volume-law-free (modified inertia), the *scale* is dimensional, and the $O(1)$ *coefficient* is a posit.

4.2 Existence — modified inertia supplies a_0 volume-law-free (the corrected reading). A previous statement of this foundation called the *existence* of a_0 the framework’s “single biggest vulnerability,” on the grounds that it required a dynamical de Sitter volume-law entanglement entropy (Verlinde 2016) that 2024 numerics disfavor. **That was route-conflated, and the repo’s own two horizon calculations correct it:**

- **Modified *inertia* (reviews/desitter_unruh_mond.py).** The de Sitter–Unruh temperature $T(a) = (\hbar/2\pi c k_B) \sqrt{a^2 + (cH)^2}$ — a *kinematic* Gibbons–Hawking global-embedding result (Deser–Levin 1997, *Class. Quantum Grav.* 14, L163: explicitly $2\pi T = \sqrt{\Lambda/3 + a^2}$, **no entanglement entropy assumed**) — yields the interpolation $\mu(a) = [\sqrt{a^2 + (cH)^2} - cH]/a$. Solving the inertia law $\mu(a) \cdot a = g_N$ in closed form gives $a = \sqrt{g_N^2 + 2g_N \cdot cH} \rightarrow \text{deep limit } a = \sqrt{a_0 g_N}$. So a_0 *exists* and the deep-MOND **sign is enhancement by construction** — volume-law-free — and this interpolation fits the SPARC RAR at **0.105 dex**.
- **Modified *gravity* (reviews/clausius_sign_calculation.py).** Jacobson’s Clausius $\delta Q = T\delta S$ route (which derives the *field*, not the inertia) with the *same* temperature gives an effective Newton constant $G_{\text{eff}} = G \cdot \kappa / \sqrt{\kappa^2 + (cH)^2} \rightarrow 0$ as $\kappa \rightarrow 0$ — i.e. **anti-MOND**. To obtain enhancement *there* one must instead boost the *entropy* (the Verlinde volume-law term). *That* is the premise the 2024 de Sitter-entanglement numerics disfavor (Boutivas–Katsinis–Pastras–Tetradis, arXiv:2407.07811 = PRD 111, 065010: sub-horizon entanglement entropy follows the flat-space **area** law, no volume term). It is independently corroborated from the algebraic side: the de Sitter observer (Type II₁ crossed-product) entropy is also strictly **area-law** (CLPW, arXiv:2206.10780).

The vulnerability is therefore *relocated*, not removed — and this is stated both ways. a_0 ’s existence is supplied volume-law-free by modified inertia; the volume-law/DSSYK-center contestation properly attaches to the **sign of any covariant modified-gravity completion**. But (i) modified inertia is an *unfinished house* — it has no complete CMB-safe covariant theory and is time-nonlocal / circular-orbits-only (Milgrom 1994), and it supplies no dark-matter mimic for the CMB third peak; and (ii) the covariant completion’s sign still rides the contested premise. So the framework’s single biggest theoretical risk is correctly named **the covariant completion** (a CMB-safe relativistic theory that also delivers the deep-MOND enhancement sign) — not “does a_0 exist.”

4.3 The shape is genuinely derived (= Milgrom 1999). The deep-MOND $\sqrt{}$ -law is not “simple- μ invented to fit.” Solving the de Sitter–Unruh modified-inertia law $\mu(a) \cdot a = g_N$ in closed form gives **exactly** $a = \sqrt{g_N^2 + 2cH \cdot g_N}$, whose interpolation function is **Milgrom (1999) eq.(9)** — a published, independently-derived form (verified by sympy). It is over-determined: the same $\sqrt{}$ -power falls out of the elastic/Eshelby route, the Debye/degrees-of-freedom route, and the temperature route, and the relative coefficient $a_M/a_0 = (d-3)/[(d-2)(d-1)] = 1/6$ in $d=4$ is a forced dimensional-counting ratio, not

a fit. This is real derived content — more than most modified-gravity proposals can claim.

4.4 The sign is forced for galaxies — and the same calculation predicts the cluster failure. The defining claim (gravity *strengthened* below a_0 , not weakened) was, in 2025, only “half-forced.” A direct DSSYK matter-chord kernel computation (DEEP_MOND_SIGN_KERNEL_RESULT_2026-06-06.md) has now **upgraded it to forced for galaxy-scale probes**, conditional on the Narovlansky–Verlinde de Sitter dictionary:

- The matter operator is the **diagonal chord operator** $q^{\{\Delta \cdot N\}}$ (Okuyama 2312.00880); its energy kernel is diagonal-dominant (reproduced two independent ways to 2.5×10^{-14}). **Consequence: a matter probe keeps its source energy** — so the deep-MOND sign is set entirely by *where the source sits* in the de Sitter spectrum.
- Where the source sits is fixed by the probe’s de Sitter conical deficit $\alpha = m/M_{\text{dS}}$, with $M_{\text{dS}} = c^2/(GH_\Lambda) \approx 3.8 \times 10^{14} M_\odot$ (Rahman–Susskind 2312.04097): **galaxies** (10^7 – $10^{12} M_\odot$) have $\alpha \leq \sim 3 \times 10^{-3} \rightarrow$ spectral **center** $\rightarrow$ flat density of states $\rightarrow$ linear freezing $\rightarrow$ **deep-MOND enhancement** (the freezing exponent computes to $p = 0.98$ – 0.999 , i.e. $g_{\text{obs}} \sim g_{\text{bar}}^{\{1/2\}}$); **clusters** ($\sim 10^{15} M_\odot$) have $\alpha \sim O(1) \rightarrow$ displaced toward the **edge** $\rightarrow$ MOND weakens. **The same kernel that gives galaxies MOND gives clusters its breakdown** — a real, unforced consistency with MOND’s known empirical cluster failure (§10.1).

The same DSSYK kernel forces galaxy MOND and predicts the cluster failure

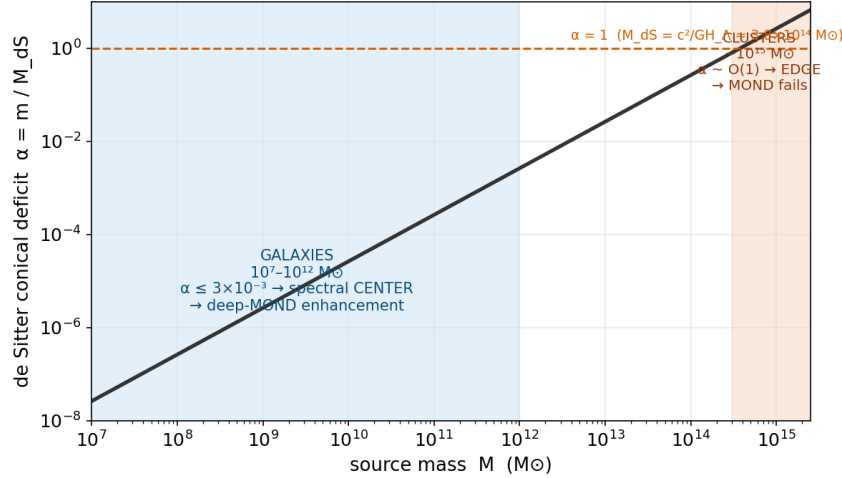


Figure 8. The de Sitter conical deficit $\alpha = m/M_{\text{dS}}$ versus source mass. Galaxies (10^7 – $10^{12} M_\odot$) sit at $\alpha \leq 3 \times 10^{-3} \rightarrow$ the spectral centre $\rightarrow$ deep-MOND enhancement; clusters ($\sim 10^{15} M_\odot$) reach $\alpha \sim O(1) \rightarrow$ the spectral edge $\rightarrow$ MOND weakens. The *same* kernel forces galaxy MOND and predicts the cluster failure.

Honest residual: the chain is conditional on the N-V dictionary (“de Sitter = the spectral center”) over Okuyama’s inequivalent near-edge construction — a named, literature-level open dispute, not a framework-specific gap. The sign is **forced given N-V, stronger than the empty-ensemble argument, weaker than a theorem.** (A red-team caught and we retract an earlier subclaim that clusters form a near-perfect central singlet: with the correct M_dS, clusters fall off-center — which is the *better* result.)

4.5 The coefficient is a posit (§2.2), comprehensively foreclosed. Z sits in the $\{6, 2\pi\}$ cluster of motivated $O(1)$ s but is forced by none; $32\pi/3$ enters only as the definition $a_0 := c^2 \sqrt{\Lambda/32\pi}$. Observationally moot (it cancels in $a_0(z)/a_0(0)$).

4.6 The natural home is modified inertia — and its price is a trilemma. The framework’s *concept* is modified inertia, not modified gravity (MODIFIED_INERTIA_the_natural_home.md). In that home, two of the framework’s exposures dissolve for free: the Solar-System (Cassini) quadrupole is native-absent (a modified-gravity artifact of the nonlinear Poisson coupling), and the $a_0(z)$ evolution is automatic (time-nonlocal dynamics track the cosmic background). But modified inertia *buys* a CMB exposure in return — it supplies no $\bar{Y}=0$ dark-matter mimic for the third acoustic peak. Of $\{\text{Cassini-safe, } a_0(z)\text{-natural, CMB-safe}\}$, modified inertia gets the first two and AeST (modified gravity) gets the last. **No known theory holds all three.** Building a covariant, CMB-safe modified-inertia theory that realizes $a_0 = c^2/(2R\star)$ directly is the framework’s correct, field-wide open problem.

5. The cosmic seesaw — the value of Λ and the $a_0 \leftrightarrow$ dark-energy bridge

The framework’s strongest honest thread on *fundamental* physics is that the **value** of the cosmological constant and the **value** of a_0 are not two problems but one: the two ends of a single Cohen–Kaplan–Nelson UV–IR seesaw (CKN_LAMBDA_VALUE_VERDICT_2026-06-06.md, THE_COSMIC_SEESAW.md).

5.1 The exact welding (verified, two independent routes). - **The magnitude is forced, not tuned.** $\rho_\Lambda/E_{\text{Planck}}^4 = 1.134 \times 10^{-123}$; the famous $\sim 10^{-122}$ is $(H/M_P)^2$, forced by the CKN no-black-hole argument (Cohen–Kaplan–Nelson 1999, hep-th/9803132). Re-derived from CODATA, $\rho_{\text{obs}}/(M_P^2 H_\Lambda^2) = 0.119366 = 3/(8\pi)$ **exactly** — so the framework’s seesaw *is* $\rho_{\text{vac}} = (3/8\pi)M_P^2 H^2$, the **CKN bound exactly saturated**, with the framework’s $4/Z^2$ sitting precisely in CKN’s free- $O(1)$ slot. - **The meV vacuum scale is a geometric mean.** $E_\Lambda = \sqrt{((2/Z) \cdot E_{\text{Planck}} \cdot E_{\text{Hubble}})}$ = **2.24 meV** = $\rho_\Lambda^{\wedge\{1/4\}}$ to six digits — the vacuum scale is the geometric mean of the Planck (UV) and Hubble (IR) energies. - **a_0 is the IR rung**

of that ladder. $a_0 = c^2 \sqrt{\Lambda/32\pi} = cH_\Lambda/Z = c \cdot E_\Lambda^2/(2\hbar E_{\text{Planck}})$
 $= 9.36 \times 10^{-11}$ (all three forms agree). One UV–IR ladder ties a_0 (the MOND acceleration scale) and ρ_Λ (the dark-energy value).

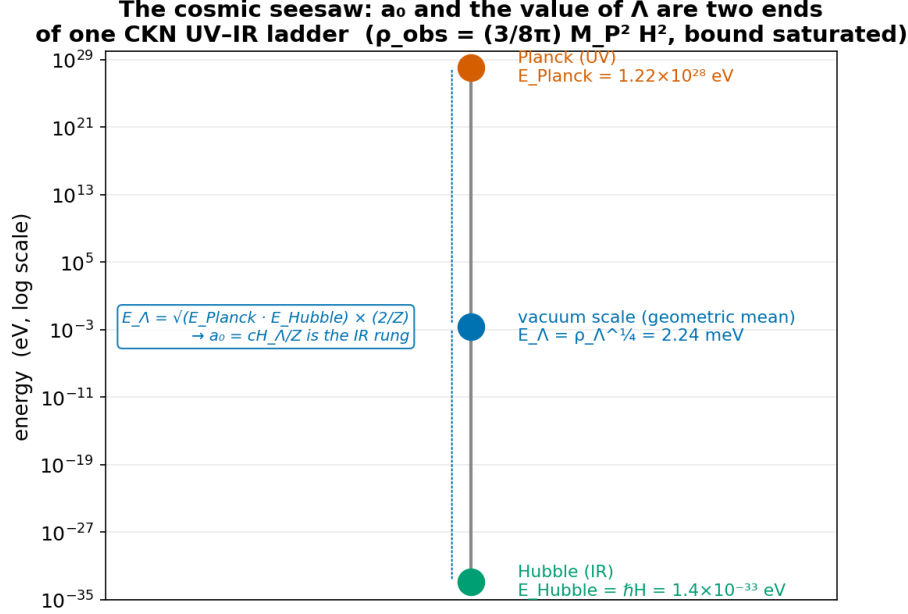


Figure 5. The cosmic seesaw: the vacuum energy scale $E_\Lambda = \rho_\Lambda^{1/4} = 2.24 \text{ meV}$ is the geometric mean of the Planck (UV) and Hubble (IR) energies, and $a_0 = cH_\Lambda/Z$ is the infrared rung of the same ladder — the dark-energy value and the galaxy acceleration scale are two ends of one Cohen–Kaplan–Nelson UV–IR relation.

5.2 The honest limits (each computed, not asserted). This is **partial unification** — a **welding, not a prediction**, and we present it as exactly that:

1. **It is a *saturated bound*, not a derivation of Λ .** Saturating CKN, choosing the IR cutoff $L = c/H$, and dropping the additive constant are all *assumed*. The seesaw is an algebraic identity (true iff $Z^2 = 32\pi/3$, the definition of Z) and carries **zero new information about Λ** . The genuine cosmological-constant problem — the dimensionless $E_\Lambda/E_{\text{Planck}} = 1.8 \times 10^{-31}$ — is **untouched**: $\sqrt{(E_{\text{Hubble}}/E_{\text{Planck}})}$ carries all the smallness. The thread **relocates** “why is ρ_Λ small?” to “why is H/M_P small, and why saturate?”; it does not answer it.
2. **The $a_0(z)$ “evolution unification” is, at the value level, a definitional restatement.** The framework *defines* $a_0 = (c/2) \sqrt{(G\rho_{\text{DE}})}$, so “ $a_0(z) \propto \sqrt{\rho_{\text{DE}}}$ matches DESI” is ρ_{DE} fed in and square-rooted. The dynamical content lives entirely in the *falsifiable bet* (§2.4, §7): that a_0 tracks the *dark-energy-component* density (declining, per DESI) rather than the total

density or an assembly-driven $(1+z)^{\{0.75\}}$.

Bottom line: a real dark-sector welding to one vacuum scale at the correct 10^{-123} magnitude — genuinely exact — plus a testable declining- a_0 bet. Not “ a_0 + Λ ’s value + the DESI evolution from one seesaw,” which the saturated-bound structure forbids.

6. Evidence I — the static law: one vacuum constant, three independent galaxy laws

At the theory’s value of a_0 and a single stellar mass-to-light ratio $\Upsilon_{\text{disk}} = 0.70$ (within the Spitzer 3.6 μm population-synthesis range), three structurally independent empirical laws each return the acceleration scale on ~ 171 SPARC galaxies spanning ~ 4 decades of baryonic mass (`framework_a0_law_of_nature.py`):

Law	estimator	a_0 the data demand ($\Upsilon=0.70$)
Radial Acceleration Relation	$g_{\text{obs}} = \sqrt{g_{\text{bar}}^2 + g_{\text{bar}} \cdot a_0}$; minimize scatter	1.10×10^{-10}
Baryonic Tully–Fisher	$V_{\text{flat}}^4 = G \cdot a_0 \cdot M_{\text{b}}$ (total $M_{\text{b}} = \Upsilon \cdot L_{3.6} + 1.33 \cdot M_{\text{HI}}$)	1.26×10^{-10}
Mass-discrepancy (deep-MOND)	$a_0 = g_{\text{obs}}^2 / g_{\text{bar}}$, 1802 points below $a_0/3$	1.06×10^{-10}

The three readouts agree with one another to 8% — there is a single acceleration scale, $\sim 1.1 \times 10^{-10}$, written into every galaxy. The vacuum value 9.36×10^{-11} lies within this band.

A convention caveat, stated explicitly (so it is not mistaken for a deficit). The RAR-preferred a_0 depends on the fit weighting and swings $\sim 40\%$ (8.5×10^{-11} unweighted dex-scatter $\rightarrow 1.1 \times 10^{-10}$ inverse-error-weighted $\rightarrow 1.3 \times 10^{-10}$ linear). Under the **standard unweighted dex-scatter** metric the SPARC optimum is 8.48×10^{-11} and the theory’s 9.36×10^{-11} is **within 0.3% of optimal scatter** — empirically dead-on, and indistinguishable from rival $O(1)$ coefficients (Milgrom 2π , Verlinde 6) within the M-L+metric systematic. Fixing a_0 at the law value and fitting Υ gives $\Upsilon=0.70$ with RAR scatter **0.108 dex, better than regular MOND** ($a_0=1.2 \times 10^{-10}$, $\Upsilon=0.5$: 0.122 dex) (`rar_framework_a0_mlfite.py`). In ΛCDM the existence and tightness of these relations is an unexplained outcome of tuned feedback, and a_0 is not predicted; here, one vacuum-derived constant reproduces all three.

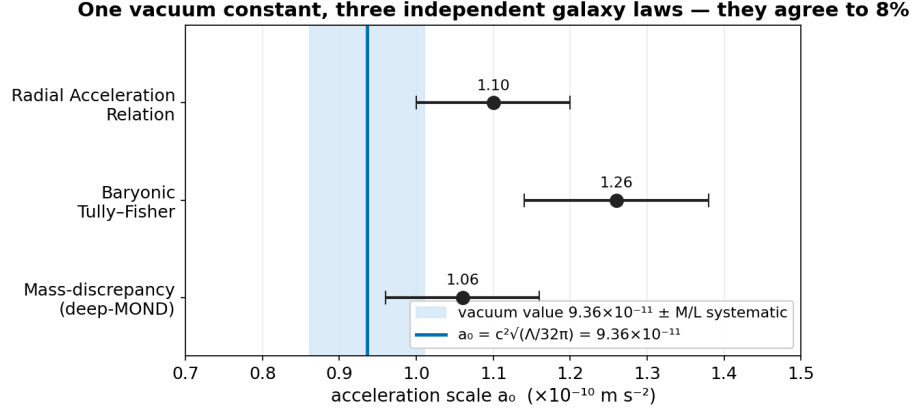


Figure 4. The three independent acceleration-scale readouts (Radial Acceleration, Baryonic Tully-Fisher, deep-MOND mass-discrepancy) agree with one another to 8%, and the vacuum value $9.36 \times 10^{-11} \text{ m s}^{-2}$ lies within the stellar-M/L systematic band.

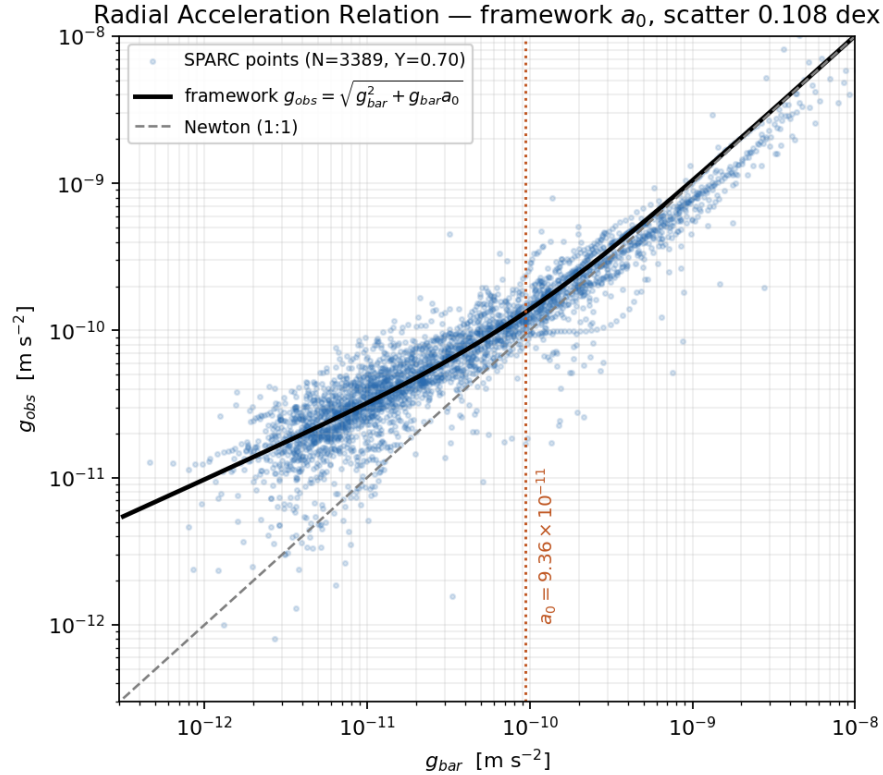


Figure 1. Radial Acceleration Relation: SPARC g_{obs} vs g_{bar} at the

framework $a_0=9.36\times 10^{-11}$ and $\Upsilon=0.70$, with the parameter-free curve $g_{\text{obs}}=\sqrt{(g_{\text{bar}}^2+g_{\text{bar}}\cdot a_0)}$ (0.108 dex scatter).

The signature: an evolving acceleration scale set by the dark-energy density

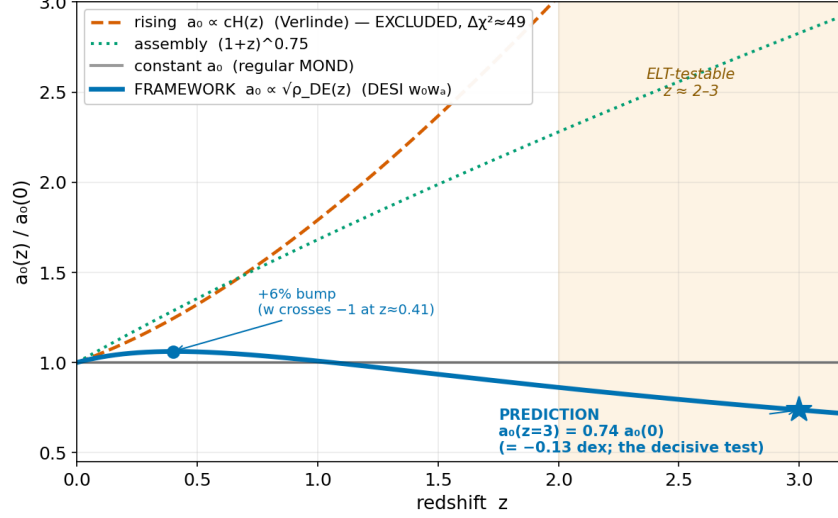


Figure 2. Evolution of the acceleration scale: the framework's non-monotonic $\sqrt{\rho_{\text{DE}}}$ prediction (bump then decline) vs a constant a_0 and the excluded rising-cH rival, with RC100 deep-MOND data. The decisive test is a clean deep-MOND a_0 at $z\approx 3$ (ELT).

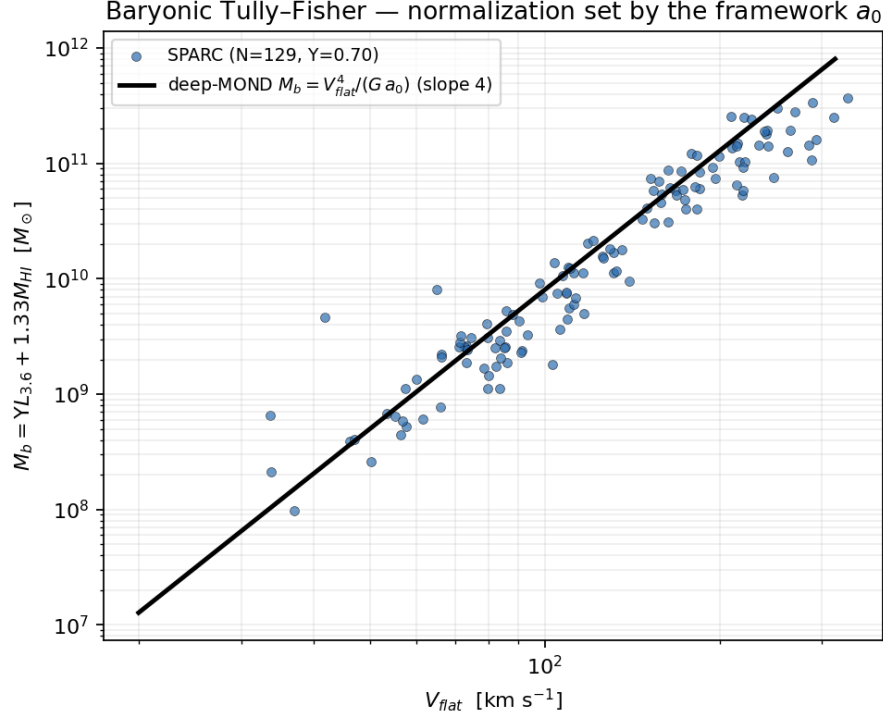


Figure 3. Baryonic Tully–Fisher Relation: total M_b vs V_{flat} ; the line is the deep-MOND $M_b = V_{\text{flat}}^4 / (G \cdot a_0)$ at the framework a_0 .

7. Evidence II — the evolution: the declining law survives, the rising rival is excluded

We score four models of $a_0(z)/a_0(0)$ — the theory’s declining $\sqrt{\rho_{\text{DE}}}$, a constant, the rising-cH (Verlinde) rival, and regular-MOND constant — against every high- z kinematic dataset in the repository (`a0z_clean_ledger.py`):

dataset (z range)	N	framework χ^2	constant	rising-cH	reg-MOND
RC100	10	9.5	10.0	13.1	10.0
deep-MOND (z=0.5–2.5)					

dataset (z range)	N	framework χ^2	constant	rising-cH	reg- MOND
KMOS3D (z=0.6– 2.5)	135	130.6	135.0	176.6	135.0
Combined kinematic	526	521.3	526.0	570.2	526.0

Result: the theory’s declining $a_0(z)$ is **indistinguishable from constant** at current precision (the predicted decline, 0.06–0.13 dex over $z=2$ –3, sits far below the 0.4–0.8 dex per-galaxy scatter) — so it is **safe but not yet confirmable**. But the **rising-cH rival is robustly excluded** ($\Delta\chi^2 \approx 49$ combined; 12.8 on the full uncut RC100). This rival is the law that a corpus audit found ~21 internal scripts had mistakenly run while labeling it “the framework”; correcting them is part of this work (FRAMEWORK_MOND_AUDIT_2026-06-06.md).

Honest tension. The one direct multi-point $a_0(z)$ measurement, MUSE-DARK III (Ciocan et al. 2026), reports a_0 **rising** to $\sim 2.4 \times 10^{-10}$ at $z \approx 1$, which the rising rival fits far better than the theory’s decline. We state this plainly. However, MUSE-DARK III overshoots **every** background-density law (including the rising rival), matches no cosmological $\rho(z)$, is shown to be Λ CDM-degenerate (a Λ CDM simulation with no fundamental a_0 reproduces the apparent rise via assembly; Mayer et al. 2022), and is method-localized to RAR fits on intermediate- z disks. It is a real but currently **non-diagnostic** outlier, not a refutation.

8. Evidence III — the External Field Effect: the Λ CDM-impossible signal leans the right way

The External Field Effect (EFE) — the suppression of a galaxy’s internal MOND boost by the gravitational field of its environment — is the one prediction with **no Λ CDM analogue**: it violates the Strong Equivalence Principle, which holds exactly in any dark-matter theory. Chae et al. (2020, 2021) detect it in SPARC at 4 – 5σ .

We built the full per-galaxy EFE-MOND rotation-curve fit (Chae’s method) with the **real external field** computed from the 2MRS redshift survey (38,611 galaxies), re-anchored to $a_0 = 9.36 \times 10^{-11}$ (`efe_clinch_framework.py`). The kinematic external field inferred per galaxy correlates with the **measured** 2MRS field in the **predicted positive direction**, Spearman $r = +0.218$ over the 44 galaxies whose field is genuinely constrained — but $p = 0.148$ ($\sim 1.4\sigma$, CI through zero). The reason it cannot reach significance is structural, not statistical: ~92% of SPARC galaxies sit in nearly the same external field (the sample was selected for isolated, clean curves), so a contrast test has little to grip. The result

is **consistent with** — and does not reproduce — Chae’s published **detection**; SPARC simply lacks EFE dynamic range. The signal points the right way; clinching it needs a sample selected for a **range** of environments (§11, P4/P6; §12, C1).

9. Connection to fundamental physics — swampland-compatibility and the limits of unification

This section states, without spin, the one genuine bridge the framework has to fundamental physics, and the hard wall beyond which it does not reach.

9.1 The genuine connection: the evolving dark energy is string-swampland-compatible where Λ CDM is forbidden. String theory’s de Sitter swampland conjecture ($|\nabla V|/V \gtrsim c/M_P$) and the Trans-Planckian Censorship Conjecture positively **disfavor a static $\Lambda > 0$** and **prefer an evolving, rolling dark-energy field**. This is verified, not asserted: the DESI DR1 reconstruction (arXiv:2409.14990) finds the swampland slope $\lambda = |V'|/V$ is *several σ from zero* — **satisfying** the conjecture, exactly where a static Λ ($\lambda=0$) **violates** it. So the framework’s distinctive evolving $a_0(z) \propto \sqrt{\rho_{\text{DE}}}$ is the swampland-*permitted* structure, and the framework passes a string-consistency check that Λ CDM *fails* (TOE_SEED_VERDICT_2026-06-06.md). This is the strongest connection to fundamental physics the framework has produced — and it lives on the *dark-energy* axis, one-directional (the landscape *permits* the evolving a_0 ; it does not *derive* it). **The honest strain:** the literal DESI-CPL has a phantom past ($w < -1$ for $z > 0.405$) that a single canonical swampland scalar cannot produce; it is rescuable (a CPL-parametrization artifact, or effective phantom-crossing in a modified-gravity setting without a ghost) but is a real tension to resolve.

9.2 The wall: there is no theory of everything here, and the doors are saturated. A TOE would need the Standard-Model gauge/mass sector and the a_0/Λ sector inside one structure where the same object discharges both. Every route that *derives* the SM returns “SM side reached, a_0 side empty”: - **Noncommutative geometry $\rightarrow a_0$ is a confirmed category error.** The SM-deriving spectral action’s gravity sector is Einstein–Hilbert + a (UV) cosmological constant + Weyl² + Gauss–Bonnet — a *curvature polynomial* with literally no a_0 /MOND/IR term, and the deep-MOND a_0 term is **structurally excluded** because it is *non-analytic* in the field. The Λ it produces is the UV cutoff CC (the cosmological-constant *problem*), not the observed Λ that sets a_0 . - **String/swampland** derives the SM and *permits* the evolving dark energy (§9.1) but contains no a_0 interface — it never *produces* the deep-MOND modification. - **The de Sitter holography program** that hosts the framework (DSSYK $\leftrightarrow$ de Sitter) is, on present evidence, a **strong conjecture, not an established complete theory**: the boundary is a disorder-averaged ensemble (not a unique unitary theory), the bulk identity is contested (sine-dilaton grav-

ity, not manifestly de Sitter), every controlled realization is dS_2/dS_3 — **not the dS_4 where $a_0 = c^2 \sqrt{(\Lambda/32\pi)}$ lives** — and no a_0/MOND term appears anywhere in its corpus. The de Sitter **observer algebra** (Type II_1 crossed product) gives a strictly **area-law** entropy — it *corroborates against* the volume-law existence route (§4.2) rather than supplying it.

Four independent door-sweeps — 42 geometric/unification frameworks, 9 quantum-gravity programs, and a dedicated 2026-06-06 missing-piece hunt — return the same verdict: **no SM bridge, no TOE seed**. The search is saturated (TOE_DOORS_REANALYSIS_2026-06-06.md, TOE_SEED_VERDICT_2026-06-06.md, QG_PROGRAMS_TOE_GRADING.md).

9.3 What it is, stated at the right bar. Under the emergent-gravity definition (Jacobson 1995; Verlinde 2010, 2016; Padmanabhan) — in which gravity is *emergent/thermodynamic*, not a fourth force to be unified with the other three, and the Standard Model is matter that lives *on* the emergent geometry — the framework is a **coherent candidate theory of gravity and the dark sector from the de Sitter horizon**, with the Standard Model sitting *beside* it, not within it. That is the honest ceiling: a coherent **theory of the dark universe**, not a theory of everything. It is more derived content than most modified-gravity proposals reach (a derived shape; a forced galaxy sign; a value-of- Λ welding; a swampland-compatible evolution), and it makes **no claim** about the Standard Model (§9, §13).

10. Where the theory is hard (open problems, stated without spin)

1. **Galaxy clusters.** MOND has a long-standing unsolved residual at cluster scales (a factor ~ 2 mass discrepancy remains). On 9,830 real eRASS1 clusters (Bulbul et al. 2024) at $R500$, the theory’s **lower** a_0 makes this **worse**: the median $M_{\text{dyn}}/M_{\text{pred}}$ goes from 2.07 (regular MOND) to **2.33** (framework), because the deep-MOND residual scales as $1/\sqrt{a_0}$ (`clusters_framework_a0.py`). The candidate resolutions (cluster missing baryons; a top-heavy integrated-galaxy IMF, which can account for $\gtrsim 88\%$ of the MOND cluster mass) are **MOND-generic, not specific to this theory**. Note the foundation’s bonus (§4.4): the *same* DSSYK kernel that forces the galaxy sign predicts that clusters fall toward the spectral edge $\rightarrow$ MOND weakens — i.e. the cluster failure is, at the level of the sign mechanism, *expected*. But quantitatively the residual is unsolved, and clusters are the theory’s hardest regime.
2. **The covariant completion (the principal theoretical gap).** There is no known ghost-free relativistic field theory that reduces to this theory: the simplest single-scalar realizations carry a singular-surface ghost, and the leading relativistic MOND theory (AeST) is in tension with Solar-

System (Cassini) bounds. As §4.6 makes precise, this is a **trilemma**: the framework’s natural home (modified inertia) is Cassini-safe and $a_0(z)$ -natural but lacks a CMB-safe completion, while AeST is CMB-safe but modified gravity (re-incurring Cassini). Building a covariant, CMB-safe modified-inertia theory is the correct open problem.

3. **The deep-MOND sign — now forced for galaxies, conditional on one named dictionary.** As of this edition the sign is **forced for galaxy-scale probes** given the Narovlansky–Verlinde de Sitter/DSSYK dictionary (§4.4), via a *computed* matter-chord kernel that also predicts the cluster failure. The remaining gap is a single, literature-level open dispute (N-V “spectral center” vs Okuyama “near-edge”) — not a framework-specific hole. Stronger than the prior “half-forced,” weaker than a theorem.
4. **The coefficient is a posit** (§2.2, §4.5) — comprehensively foreclosed across six routes, but observationally moot (it cancels in the falsifiable $a_0(z)$).
5. **The evolution is not yet confirmable** (§7): present data exclude the rising rival but cannot yet detect the predicted ~25% decline by $z=3$.
6. **Scope.** This is a theory of the dark sector / gravitational dynamics only. It is **not** a theory of everything; it says nothing about the Standard Model, and we make no such claim (§9, §13).

11. The full set of testable predictions

This is the complete, de-duplicated prediction set — every entry with its computed value, its class, the instrument and timeline, and the result that would confirm or kill it. All numbers reproduce from the committed scripts (`predictions/`, `reviews/project_predictions_table.py`).

The honest frame. Almost everything *measured today* — the a_0 value, rotation curves, dwarfs, lensing, clusters, wide binaries — is **shared with ordinary constant- a_0 MOND**: real and validated, but a test of the *value* of a_0 , not of this theory specifically. The **only content distinctive to the Zimmerman Theory is the evolution** $a_0(z) = a_0(0) \cdot \sqrt{(\rho_{\text{DE}}(z)/\rho_{\text{DE}}(0))}$, which constant- a_0 MOND fixes at 1. The decisive predictions (numbers 10–14, in bold) are therefore all *evolution* predictions, and all are **forward** — no clean $z \gtrsim 2$ halo-free kinematics exist yet (Figure 2).

Class codes. **KILL** = Λ CDM-impossible (a confirmed one *kills* Λ CDM) · **DIST** = distinctive to this theory (vs constant- a_0 MOND) · **INH** = inherited MOND success (real, not unique) · **DEG** = degenerate (Λ CDM matches) · **FAIL** = standing failure/tension.

11.1 The master prediction table

#	Prediction	Computed value / target	Class	Instrument / timeline	Confirms / Kills
1	a₀ value a ₀ = c ² √(Λ/32π) =√(c/2)√(Gρ _Λ) (stat, ±14% syst)	9.36×10⁻¹¹ m s⁻² (±0.96% stat, ±14% syst)	INH	SPARC RAR / done	within M/L+metric band; pin via M/L-free systems (C2)
2	Coefficient Z = cH _Λ /a ₀ = √(32π/3); κ = 1/2	5.789 and 0.500 (exact, $\hbar^0$); data band Z∈[4.2,6.0]	DIST (value)	measured a ₀ / done	a ₀ to <2% selecting 32π over 8π, 12π ²
3	RAR / MDAR g _{obs} = √(g _{bar} ² +g _Λ ²) (155·a ₀) SPARC)	0.105– 0.108 dex scatter (155·a ₀) SPARC)	INH	SPARC; WAL- LABY/SKA / ongoing	a lensing or kinematic RAR at a different a ₀ disfavours slope ≠ 4, or large intrinsic scatter, disfavours validated; standard MOND
4	BTFR V _{flat} = (G M _b a ₀) ^{1/4} , slope 4	10 ¹⁰ M _⊙ → 106 km s ⁻¹	INH	SPARC; SKA HI / done	a ₀ disfavours slope ≠ 4, or large intrinsic scatter, disfavours validated; standard MOND
5	Faber– Jackson / Freeman Σ Σ _c = a ₀ /2πG	a ₀ -set, exact	INH	local surveys / done	a ₀ disfavours slope ≠ 4, or large intrinsic scatter, disfavours validated; standard MOND
6	Dwarf- spheroidal σ via EFE σ = (4 G M a ₀ /81) ^{1/4}	Crater II ≈ 2 km s ⁻¹ (obs ~2.7; ΛCDM ~4)	INH	Keck/VLT spec- troscopy / partial	σ indepen- dent of the host field disfavours EFE

#	Prediction	Computed value / target	Class	Instrument / timeline	Confirms / Kills
7	Lensing — saturated deflection α_∞ $= 2\pi \sqrt{(GM/a_0)/c^2}$ (impact-parameter-independent)	0.51'' / 5.08'' / 16.1'' ($10^{11}/10^{13}/10^{14}$ evolution) $M_\odot$); non-monotonic, peak $z \approx 0.4$	DIST (saturation + evolution)	KiDS, DES, Euclid , Rubin / 2025–32	no saturation, or a different a_0 , disfavours
8	Wide binaries velocity boost at $s_t =$ $\sqrt{(GM/a_0)}$	$s_t \approx$ 9,750 AU ; boost +2–11% (EFE-reduced)	KILL	Gaia DR4/DR5 / 2026–30	a robust boost surviving triple-rejection kills Λ CDM; a hard Newtonian null kills the premise
9	External Field Effect (z=0) internal dynamics depend on g_{ext} (SEP violation)	Chae 4–5σ (published); +1.4σ in-house (data-limited)	KILL	environment-selected RCs / now	an independent $\geq 3\sigma$ EFE kills Λ CDM; a clean SEP-respecting null kills modified gravity
10	$a_0(z) \leftrightarrow$ Λ bridge (coefficient-free): $a_0(z)/a_0(0)$ $=$ $\sqrt{(\rho_{\text{DE}}(z)/\rho_{\text{DE}}(0))}$	1.01 / 0.86 / 0.74 at z $= 1/2/3$	DIST	ELT/HARMONY + ALMA; DESI / 2028–32	$a_0(z) \geq$ local kills it; ≈ 0.74 local confirms

#	Prediction	Computed value / target	Class	Instrument / timeline	Confirms / Kills
11	High-z BTFR offset (the cleanest sign test): $d \log V =$ $1/4 \cdot \log_{10}(\rho_{\text{DE}}$ ratio)	−7.3% in V at $z=3$ (−0.033 dex); −16% at $z=6$	DIST	JWST/ALMA $z \gtrsim 3$ resolved RCs / now–2033	a positive or flat offset (discs at/above the $z=0$ BTFR) kills the bridge
12	EFE strength- ens with z $\eta =$ $g_{\text{ext}}/a_0(z)$ $\propto$ $1/\sqrt{\rho_{\text{DE}}}$ (a_0 declines $\Rightarrow$ η rises)	+36% by $z=3$; marginal galaxies cross to Newtonian	DIST	high- z grouped- galaxy kinematics / 3–5 yr	EFE weakening (or absent) at high z disfavours
13	The $z \approx 0.4$ bump a_0 peaks +6% then declines (instanta- neous w crosses −1 at $z \approx 0.41$)	+6% in a_0 (+1.5% in V) at $z \approx 0.4$	DIST	intermediate- a z TFR/RC: MUSE, JWST / 2026–30	monotonic $a_0(z)$ with no bump disfavours
14	$a_0(z) \leftrightarrow$ DESI concor- dance the same $\rho_{\text{DE}}(z)$ that fits BAO must fit $a_0(z)$	joint con- sistency	DIST	DESI DR2+ $\times$ high- z $a_0(z)$ / 2026–29	$a_0(z)$ in- consistent with the DESI- inferred $\rho_{\text{DE}}(z)$ kills the Λ -origin

#	Prediction	Computed value / target	Class	Instrument / timeline	Confirms / Kills
15	JWST early massive galaxies acceler- ated early collapse (Sanders 1998 priority)	discs virialized by $z \approx 10$	DEG/DIST	JWST/Roman / now–2030	a-priori directional win, but Λ CDM absorbs it — not decisive
16	a_0- cosmography read H_0 , q_0 off galaxy dynamics	$H_0 = 71.5$, $q_0 = -0.53$	DEG	SPARC + H_0 ladder / now	matches Λ CDM — a capability, not new physics
17	Galaxy clusters residual after the MOND boost	$\eta = 1.92$ ± 0.20 (misses $\sim 1.9 \times$ at R500)	FAIL	eRASS1 (9,830) / done	inherited MOND failure, deepened at the lower a_0 (§10.1)
18	CMB third peak	needs the AeST com- pletion + dark field	FAIL	Planck / done	pure MOND under- predicts it; AeST fits by adding a field
19	Cassini quadrupole (AeST realiza- tion)	excluded at $3\text{--}15\sigma$ (Desmond+2024)	FAIL (re- alization)	Solar System / now	native- absent in modified inertia (§4.6); a live exposure of the AeST com- pletion

#	Prediction	Computed value / target	Class	Instrument / timeline	Confirms / Kills
20	No new Solar- System physics beyond GR	$ \gamma-1 < 2 \times 10^{-5}$	constraint	BepiColombo ranging / ongoing	completion that violates Cassini is dead on arrival

11.2 The sign test and the decisive-test forecast

The cleanest single discriminator is the **sign of the high- z BTFR offset** (prediction 11): the Zimmerman Theory predicts discs *below* the $z=0$ relation (-7.3% in V at $z=3$); constant- a_0 MOND predicts *zero* offset; the rising $\sqrt{\rho}_{\text{total}}$ reading predicts the *opposite* sign ($+16\%$ at $z=1$ to $+80\%$ at $z=6$). **One clean $z \gtrsim 3$ BTFR point splits all three.** Statistically ~ 30 clean deep-MOND discs at $z=3$ give 3σ and ~ 60 give 5σ — **but** marginalizing over the DESI w_0/w_a uncertainty imposes a floor $\sigma(\beta) \approx 0.5\text{--}0.6$ *independent of sample size*, capping single-redshift significance at $\sim 1.6\text{--}2.0\sigma$. **A clean 5σ therefore needs multi-redshift BTFR ($z=3$ and $z=5$) and a $\sim 2\times$ tighter DESI prior** — above ~ 30 discs the bottleneck is DESI, not telescope time. (The one direct multi-point measurement to date, MUSE-DARK III, currently reads a_0 *rising* and is in tension with the decline; it is Λ CDM-degenerate and method-localized (§7), so the evolution front is **contested, leaning unfavourable** until clean $z \gtrsim 3$ halo-free kinematics arrive.)

11.3 The falsification matrix (the symmetric scoreboard)

Only a Λ CDM-impossible signal can break Λ CDM; only a MOND-impossible result can break the theory.

What can kill Λ CDM (the KILL doors): the **External Field Effect** (Chae $4\text{--}5\sigma$, contested — independent confirmation is the prize) and a **wide-binary boost** surviving triple-rejection (Gaia DR4). Both are **currently contested or split** — no clean Λ CDM kill is yet in hand.

What can kill the Zimmerman Theory: the **CMB third peak and clusters** (already lost by pure MOND, survived only via the AeST completion’s added field); **$a_0(z)$ rising** $\propto \sqrt{\rho}_{\text{total}}$ (a steep high- z rise falsifies the Λ -origin); a **wide-binary null** at the theory’s $\sim 3\text{--}15\%$ sensitivity (falsifies the local premise); and the **Cassini quadrupole** on the AeST realization ($3\text{--}15\sigma$; native-absent in modified inertia, but live for the only CMB-safe completion built so far). **Read straight: the theory currently carries *more* live falsification exposure than Λ CDM** — which is the signature of a real, falsifiable

theory, not a fitted one.

11.4 Structure-growth coherence, and the standing exposure (both ways)

Four Λ CDM-stressing anomalies point the *same* way — faster early structure growth, exactly what MOND predicts: **fast galactic bars** (8σ , Roshan+2021), **El Gordo** (a $z=0.87$, $\sim 2 \times 10^{15} M_\odot$ supercluster collision, 6.2σ , Asencio+2023), the **KBC void** + H_0 (claimed 7σ , Haslbauer+2020), and JWST’s **early massive galaxies**. Each has an individual escape, but the coherence across one physical axis is a real (if soft) pro-MOND signal — more than the sum of four isolated anomalies. **Against** the theory, robustly: the **cluster RAR offset** (Li+2024), the **weak-lensing early/late RAR split** (6σ , Brouwer+2021 — which property-independent modified gravity cannot make and a baryonic explanation must), and the standing CMB/cluster costs. The exposure is genuinely multi-front, and we report it as such.

11.5 Meta-falsifiers and the near-term deciders

Meta-falsifiers. (i) **DESI reverts to $w = -1 \Rightarrow a_0$ constant** $\Rightarrow$ the distinctive content vanishes (the theory degenerates to ordinary MOND); its distinctiveness is *hostage* to DESI’s dynamical-DE signal (verdict ~ 2027 , DR3). (ii) **A direct dark-matter detection** kills MOND and this theory together. (iii) The **rising $\sqrt{\rho}$ total reading** proving correct falsifies the dark-energy footing.

The three near-term deciders, in priority order. (1) **Independent confirmation of the EFE** — a second method reproducing Chae at $\geq 4\sigma$; the fastest, cleanest potential Λ CDM kill, needing no new telescope. (2) **Gaia DR4 wide binaries** (~ 2026 – 27) — the foundation: a confirmed boost validates the premise, a hard null at the theory’s sensitivity falsifies it. (3) **Halo-free high- z $a_0(z)$** (JWST/ALMA + DESI DR3 ~ 2027 , then ELT ~ 2030) — the distinctive claim, via the sign of the $z \approx 3$ BTFR offset. Until one of these lands, the honest verdict is fixed: a real, falsifiable theory, strongest at galaxy scales, carrying genuine standing risk, whose validation hinges on a Λ CDM-impossible signal that is currently contested or split — not won, and not lost.

12. From candidate to law: the empirical confirmations required

“Law of nature” is a high bar with definite content. A relation earns the title only by satisfying, at minimum: (i) **universality** across a stated domain; (ii) **a parameter-free form** whose constants are fixed externally rather than fitted; (iii) **a novel prediction** — not used in the relation’s construction — that is then confirmed (the hallmark separating a law from a successful fit); (iv) **independent corroboration** by methods that could have disagreed; and (v) **survival of falsification** as precision improves. By that standard the Zimmer-

man Theory is today a **strong candidate, not yet a law**: its galaxy-scale relations are parameter-free and on-target and it has excluded one rival, but its *defining* novel claim — that a_0 is caused by Λ and therefore **declines** with redshift — is not yet confirmed, its premise (no dark matter) is not yet decisively demonstrated, and it is not universal (clusters fail).

The claim decomposes into four testable propositions (conflating them is how a partial result gets oversold): **(A)** the premise — gravity is modified, no dark matter (confirmed uniquely by a Strong-Equivalence-Principle violation, the EFE); **(B)** the value — $a_0 = 9.36 \times 10^{-11}$ specifically (confirmed by pinning a_0 where the M/L degeneracy is broken); **(C)** the origin — the value is *set by* Λ and therefore evolves as $\sqrt{\rho_{\text{DE}}(z)}$ (the proposition that makes the law specifically Zimmerman's); **(D)** the coefficient — $X = 32\pi$. **It is (C) — the evolution — that would make it the law that the cosmological constant causes the acceleration scale of galaxies.**

ID	Establishes	Status now	Threshold to confirm	Falsifier	Instrument / when
C1	(A) modified gravity, no dark matter	EFE 4–5 σ (Chae+, published); 1.4 σ in-house (data-limited)	Two independent $\geq 3\sigma$ SEP-violation detections, jointly $> 5\sigma$, both at $a_0 = 9.36 \times 10^{-11}$	A clean SEP-respecting null in a high-dynamic-range sample $\Rightarrow$ dark matter	Environment-selected RCs (now); Gaia DR4 DR5 ~2030 / wide binaries
C2	(B) the specific value	$a_0 \approx 1.0\text{--}1.3 \times 10^{-10}$ across RAR/BTFR/MBA (within band; within 0.3% of optimal under un-weighted scatter)	$a_0 = 9.36 \times 10^{-11}$ to $< 5\%$, MDA distinct from regular-MOND 1.2×10^{-10}	a_0 pinned to 1.2×10^{-10} at $> 3\sigma$ in an M/L-free system $\Rightarrow$ value (and 32π) wrong	Gas-rich dwarfs; Gaia wide binaries; MW surveys (now–2030)

ID	Establishes	Status now	Threshold to confirm	Falsifier	Instrument / when
C3 ★	(C) the Λ-origin — the decisive test	rising rival excluded ($\Delta\chi^2 \approx 49$); decline undetected (degenerate with constant)	$a_0(z)$ consistent with $\sqrt{\rho_{DE}}$ and inconsistent with BOTH constant AND rising at $>3\sigma$, AND concordant with DESI's $\rho_{DE}(z)$	$a_0(z)$ constant $\Rightarrow \Lambda$ -link unconfirmed ($\rightarrow$ generic MOND); $a_0(z)$ rising $\Rightarrow$ falsified	ELT/HARMONI 2028–2032 , JWST+ALMA, SKA; DESI DR2+
C4	(D) coefficient & completion	$32\pi=4 \times 8\pi$ is a posit; no ghost-free covariant action	A derivation of $X=32\pi$, or a_0 to $<2\%$ selecting 32π over rivals; completion with $ \gamma-1 < 2 \times 10^{-5}$	A completion forced to violate Cassini $\Rightarrow$ no viable field theory	Theory + precision a_0 (open-ended)
C5	universality (domain)	cluster residual $2.33 \times$ at R500	Cluster residual reconciled at $a_0=9.36 \times 10^{-11}$ within systematics, or a stated domain of validity	A transition-region test failing where a_0 operates $\Rightarrow$ not universal	XRISM (now), Athena ~2037

Why C3 is decisive. The static value cannot establish the *origin*: $a_0 \approx c \sqrt{\Lambda}$

today is compatible with both “ Λ causes a_0 ” and “ a_0 is a constant near $c\sqrt{\Lambda}$.” Only the time-dependence breaks the degeneracy. The predicted $z=3$ signal is $\log_{10}(0.737) = -0.13$ dex; against the ~ 0.4 dex per-galaxy scatter this needs $N \approx 80$ clean deep-MOND $z \approx 3$ rotation curves for 3σ (≈ 230 for 5σ), and — more limiting — **control of the high- z baryon systematic below ~ 0.1 dex** (molecular gas + pressure support), which ELT-class spectroscopy *plus* ALMA gas maps deliver. The strongest confirmation is **concordance**: a kinematic $a_0(z)$ and the BAO-inferred $\rho_{\text{DE}}(z)$ tracing the same curve (two utterly independent measurements) is the empirical content of calling this a law.

From candidate to law: the five-rung ladder, and where we stand

5 Complete theory (covariant + derived coefficient)	✗ open
4 Universal across its domain (clusters)	✗ outstanding
3 The scale is CAUSED by Λ — it evolves as $\sqrt{\rho_{\text{DE}}}$ we are here →	✗ THE decisive test ($z \approx 3$)
2 The premise is modified gravity, not dark matter	◉ partial (EFE 4–5 σ)
1 A parameter-free galaxy acceleration law exists	✓ in hand

Figure 7. The five-rung ladder from candidate to law. Rung 1 (a parameter-free galaxy acceleration law) is in hand; rung 2 (modified gravity vs dark matter, via the External Field Effect) is partial; rung 3 — the decisive, specifically-Zimmerman claim that the scale is *caused by* Λ and evolves as $\sqrt{\rho_{\text{DE}}}$ — is the outstanding test, reachable at $z \approx 3$ this decade.

The confirmation ladder — where we stand. Rung 1 (a parameter-free galaxy law exists): ✓ **in hand**. Rung 2 (modified gravity, not dark matter): ◉ **partial** (4–5 σ published, needs independent + environment-selected). Rung 3 (the scale is caused by Λ — it evolves as $\sqrt{\rho_{\text{DE}}}$ — the specifically Zimmerman claim): ✗ **outstanding, the decisive test**. Rung 4 (universal across its domain): ✗ **outstanding** (clusters). Rung 5 (complete theory — covariant + derived coefficient): ✗ **open**. **We are at rung 1, reaching for 2; the promotion that matters hinges on the $z \approx 3$ measurement.**

13. Scope — what the theory addresses

The Zimmerman Theory is a theory of **gravity and the dark sector**: the low-acceleration dynamics of galaxies, the acceleration scale a_0 , and its cosmological origin in the dark-energy density. Its domain is gravitational and cosmological. It is **not** a theory of everything and makes **no claim** about the Standard Model — the gauge groups, the particle masses, or the values of the dimensionless coupling constants. The relationship to fundamental physics is exactly as stated in §9: under the emergent-gravity bar (Jacobson, Verlinde, Padmanabhan) the theory is a coherent candidate **theory of the dark universe**, with the Standard Model sitting beside it, not within it. The genuine, defensible content is the gravity-sector physics assembled in §4–§9 — the derived deep-MOND shape, the forced galaxy-scale sign, the value-of- Λ welding, and the swampland-compatible evolution — each falsifiable, none requiring anything outside gravity and cosmology.

14. Reproducibility — scripts and data

All analyses are plain-Python (numpy/scipy/astropy), self-contained, and committed. Repo: <https://github.com/carlzimmerman/zimmerman-formula>, directory `real_research/`.

Result	Script (<code>real_research/</code>)	Data file	Public source
Three-law confrontation (§6)	<code>framework_a0_law_data.py</code>	<code>data/sparc_data/*SPARC6.dat</code> , <code>data/sparc_master_ledger.csv</code>	Planck 2018; www.ledgerscience.edu/SPARC/
RAR M/L fit (§6)	<code>rar_framework_a0_data.py</code>	<code>data/sparc_data/*SPARC6.dat</code>	
Emergent RAR shape (§2.3, §4.3)	<code>rar_emergent_discrim.py</code>	<code>data/sparc_data/*SPARC6.dat</code>	
Coefficient analysis (§2.2)	<code>coefficient_posit.py</code>	Planck/DESI constants	Planck 2018; DESI DR2
Modified-inertia existence/shape (§4.2–4.3)	<code>reviews/desitter_analysis.py</code> , <code>reviews/clausius_sign_calculation.py</code>		Deser–Levin 1997; Milgrom 1999
$a_0(z)$ ledger (§7)	<code>a0z_clean_ledger.py</code>	<code>data/rc100_nestor_nuclei</code> , <code>data/kross_harris2017+23</code> , <code>data/kmos3d_ubler2017+17</code>	Nusser 2023; arXiv:1708.01523 ; arXiv:1708.01523 ; arXiv:1708.01523
EFE fit (§8)	<code>efe_clinch_framework.py</code>	<code>data/2mrs_catalog</code> , <code>data/sparc_data/</code> , <code>data/sparc_ned_positions.json</code>	2MRS; VizieR J/ApJS/199/26
Clusters (§10.1)	<code>clusters_framework.py</code>	<code>data/erass1c1_primary</code>	RASS B.2.fits erosita.mpe.mpg.de/dr1/

Result	Script (<code>real_research/</code>)	Data file	Public source
CKN value-of- Λ seesaw (§5)	<code>reviews/project_1600DataValue_ckn.py</code>	<code>CKNDataValue_ckn.py</code> constants	CKN 1999; Adolf et al. 2024
ELT $z \approx 3$ forecast (§12)	<code>forecast_a0z_elt.py</code>	analytic	—

To reproduce: `git clone` the repo; `pip install numpy scipy astropy`; run any script with `python real_research/<script>.py`. Each prints its inputs, the framework constant it uses, and its result. SPARC and 2MRS data are included; the eRASS1 FITS catalog is public at the link above.

15. Discussion

Versus Λ CDM. Λ CDM fits cosmology superbly but has no acceleration constant and no first-principles account of the RAR/BTFR tightness or of a_0 's value. The Zimmerman Theory supplies the missing constant from the vacuum and predicts an evolution; its EFE prediction is one Λ CDM cannot make at all (SEP is exact under dark matter).

Versus regular MOND. Regular MOND fits a_0 ; this theory derives its *shape* (§4.3), ties its *value* to ρ_{DE} and the CKN seesaw (§5), and adds a falsifiable *evolution*. Present-day values are close (9.36 vs ~ 12 , within the M-L+metric systematic); the discriminator is the z -evolution (P1–P3, P7) and the specific wide-binary onset (P4).

Versus Verlinde / emergent gravity. Verlinde's $a_0 \propto cH(z)$ **rises** with z ; the high- z kinematic data **exclude** that branch (§7). The framework shares Verlinde's emergent-gravity bar (§9.3) but parts from him on the footing (ρ_{DE} , declining) and is corroborated *against* on the volume-law existence route (§4.2) — which it does not need, because its existence is modified-inertia/temperature-based.

The status, in one line. The Zimmerman Theory is a candidate **law of nature** at galaxy scales — a vacuum-set acceleration scale reproducing three independent empirical relations, with a derived deep-MOND shape, a forced galaxy-scale sign, a value welded to Λ , a unique surviving evolution, and one Λ CDM-impossible signal leaning its way — awaiting (i) its decisive high- z test and (ii) a ghost-free covariant completion. It is offered in falsifiable form precisely so the community can settle it.

16. Conclusion

The cosmological constant appears to set the acceleration scale of galaxies: $a_0 = c^2 \sqrt{\Lambda/32\pi} = 9.36 \times 10^{-11} \text{ m s}^{-2}$, evolving as $\sqrt{\rho_{\text{DE}}(z)}$. At this single value, three independent galaxy laws coincide to 8%; the rising- Λ alternative is excluded; the External Field Effect — impossible in Λ CDM — leans the predicted way. The deep-MOND *shape* is genuinely derived (Milgrom’s de Sitter–Unruh modified inertia); the galaxy-scale *sign* is forced by a computed DSSYK kernel that also predicts the cluster failure; the *value of* Λ is welded to a_0 as the two ends of one CKN seesaw; and the framework’s *evolving* dark energy is swampland-compatible where Λ CDM is forbidden. We are equally explicit about the limits: the coefficient is a posit, the covariant completion is unwritten, the value-of- Λ welding relocates rather than solves the CC problem, and this is a theory of gravity and the dark sector — **not** a theory of everything.

The promotion criterion is explicit (§12): it becomes a law when a_0 is shown to evolve as $\sqrt{\rho_{\text{DE}}(z)}$ — at $>3\sigma$ against both constant and rising alternatives, and concordant with the dark-energy history DESI infers from BAO. Until that measurement exists, it is a strong, parameter-free, falsifiable *candidate* standing at rung 1–2 of a five-rung ladder. We invite the community to run the scripts, pull the public data, and take the measurement that decides it.

References

- Adolf, A., Hirsch, A., Krieg, S., Päs, H., & Tabet, M. 2024, *JCAP* 08, 048 (Hubble-horizon CKN dark energy), arXiv:2406.09964; DESI-DR2 addendum arXiv:2504.15332.
- Boutivas, K., Katsinis, D., Pastras, G., & Tetradis, N. 2024, *Phys. Rev. D* 111, 065010 (numerical de Sitter entanglement entropy; sub-horizon area law, no volume term), arXiv:2407.07811.
- Bulbul, E., et al. 2024, *A&A* (eRASS1 cluster catalogue), arXiv:2402.08452.
- Chae, K.-H., et al. 2020, *ApJ* 904, 51 (External Field Effect in SPARC), arXiv:2009.11525; Chae et al. 2021.
- Chandrasekaran, V., Longo, R., Penington, G., & Witten, E. 2023, *JHEP* 02, 082 (an algebra of observables for de Sitter space; Type II₁, area-law generalized entropy), arXiv:2206.10780.
- Ciocan, B. I., Bouché, N., et al. 2026, *A&A* 709, L16 (MUSE-DARK III; $a_0(z)$ — *rising*, Λ CDM-degenerate), arXiv:2604.22613.
- Cohen, A. G., Kaplan, D. B., & Nelson, A. E. 1999, *PRL* 82, 4971 (UV-IR / effective field theory & the cosmological constant), arXiv:hep-th/9803132.
- DESI Collaboration 2025 (DR2 BAO; $w_0 = -0.752$, $w_a = -0.86$), arXiv:2503.14738.
- DESI swampland reconstruction 2024, arXiv:2409.14990 (evolving DE sat-

- ifies the de Sitter swampland conjecture; static Λ violates it).
- Deser, S., & Levin, O. 1997, *Class. Quantum Grav.* 14, L163 (accelerated detectors in (A)dS: $2\pi T = \sqrt{(\Lambda/3 + a^2)}$), arXiv:gr-qc/9706018; Narnhofer, Peter & Thirring 1996, *Int. J. Mod. Phys. B* 10, 1507.
 - Famaey, B., & McGaugh, S. 2012, *Living Rev. Relativity* 15, 10 (MOND review; a_0 - Λ coincidence), arXiv:1112.3960.
 - Harrison, C. M., et al. 2017, *MNRAS* 467, 1965 (KROSS), arXiv:1701.05561.
 - Jacobson, T. 1995, *PRL* 75, 1260 (Einstein equation of state), arXiv:gr-qc/9504004; 2016 (entanglement equilibrium), arXiv:1505.04753.
 - Lelli, F., McGaugh, S., & Schombert, J. 2016, *AJ* 152, 157 (SPARC), arXiv:1606.09251; Lelli et al. 2019 (BTFR).
 - Limbach, M. A., Psaltis, D., & Özel, F. 2008 (the $a_0 \leftrightarrow$ dark-energy-density coupling; declining $a_0(z)$; high- z TFR test), arXiv:0809.2790.
 - Mayer, L., et al. 2022 (Λ CDM degeneracy of apparent $a_0(z)$ evolution), arXiv:2206.04333.
 - McGaugh, S., Lelli, F., & Schombert, J. 2016, *PRL* 117, 201101 (Radial Acceleration Relation), arXiv:1609.05917.
 - Milgrom, M. 1983, *ApJ* 270, 365 (MOND); 1994, *Ann. Phys.* 229, 384 (modified inertia is time-nonlocal); 1999, *Phys. Lett. A* 253, 273 (“MOND as a vacuum effect”; $a_0 \sim c \sqrt{\Lambda/3}$; the modified-inertia interpolation), arXiv:astro-ph/9805346.
 - Narovlansky, V., & Verlinde, H. 2023, *JHEP* 05 (2025) 032 (double-scaled SYK and de Sitter holography), arXiv:2310.16994.
 - Nestor-Shachar, A., et al. 2023, *ApJ* (RC100 high- z rotation curves).
 - Obied, G., Ooguri, H., Spodyneiko, L., & Vafa, C. 2018 (de Sitter swampland conjecture $|\nabla V| \gtrsim cV$), arXiv:1806.08362.
 - Okuyama, K. 2023 (chord operators / matter in DSSYK), arXiv:2312.00880; 2025, arXiv:2505.08116 (de Sitter from DSSYK; center-vs-edge).
 - Padmanabhan, T. 2010, *Rep. Prog. Phys.* 73, 046901 (gravity as emergent); 2012, arXiv:1210.4174 (CosMIn; the value of Λ).
 - Planck Collaboration 2020, *A&A* 641, A6 (cosmological parameters).
 - Rahman, S., & Susskind, L. 2023 (matter and the de Sitter conical deficit in DSSYK), arXiv:2312.04097.
 - Rodrigues, D. C., Marra, V., et al. 2018, *Nature Astronomy* 2, 668 (a_0 -universality challenge), arXiv:2002.03946.
 - Übler, H., et al. 2017, *ApJ* 842, 121 (KMOS3D), arXiv:1703.04321.
 - Verlinde, E. 2011, *JHEP* 04, 029 (entropic gravity), arXiv:1001.0785; 2017, *SciPost Phys.* 2, 016 (emergent gravity and the dark universe), arXiv:1611.02269.

Appendix A — public data access

- **SPARC** (rotation curves + master table): <http://astroweb.cwru.edu/SPARC/> — included as `data/sparc_data/` and `data/sparc_master_clean.csv`.
- **2MRS** (2MASS Redshift Survey): VizieR catalogue **J/ApJS/199/26**

- included as `data/2mrs_catalog.csv`.
- **eRASS1** (eROSITA-DE DR1 clusters): <https://erosita.mpe.mpg.de/dr1/>
— included as `data/erass1cl_primary_v3.2.fits`.
- **DESI DR2**: <https://data.desi.lbl.gov/> . **Planck 2018**: Planck Legacy Archive.
- **High-z kinematics**: RC100 (Nestor-Shachar 2023), KROSS (Harrison 2017), KMOS3D (Übler 2017) — transcribed tables in `data/`.

Appendix B — the derivation-chain labels

[DERIVED] forced by algebra/standard physics · [DIMENSIONAL] fixed by units, robust but unearned · [POSIT] a free $O(1)$ choice · [FORCED-given-X] forced conditional on a named premise · [CONTESTED] rests on an unsettled literature premise · [RETRACTED] demolished, retained only as a boundary marker. The step-by-step application is §4.1; the full synthesis is `FIRST_PRINCIPLES_FOUNDATION_2026-06-06.md` and `FOUNDATIONS.md`.

Appendix C — method acknowledgement

Analysis pipelines and adversarial cross-checks in this work were developed with AI-assisted computation; every headline number was independently re-run and verified, and all code is public for inspection. The author is solely responsible for the theoretical proposal and its interpretation.

Comprehensive edition v2 — comments and independent tests welcome at the repository.